

SIEM

What is SIEM?

SIEM (Security Information and Event Management) is a comprehensive security management solution that provides real-time analysis of security alerts generated by applications and network hardware. It combines two key functions:

1. Security Information Management (SIM):

- Collects, aggregates, and analyses log data from various sources like firewalls, intrusion detection systems (IDS), intrusion prevention systems (IPS), antivirus software, and other security-related systems.
- Provides historical analysis, reporting, and compliance management.

2. Security Event Management (SEM):

- Monitors, correlates, and analyses events in real-time, enabling the detection of security incidents as they happen.
- Supports incident response by providing alerts and allowing for quick investigation of potential security threats.

Key Functions of a SIEM:

Log Management: Collects and centralizes logs from multiple sources for easy management and analysis.

Event Correlation: Correlates events from different sources to identify patterns or anomalies that may indicate a security threat.

Alerting: Generates alerts for suspicious activities or potential security incidents.

Incident Response: Helps security teams respond to incidents by providing detailed information and tools for investigation.

Compliance Reporting: Assists organizations in meeting regulatory compliance by generating reports based on log data.

Enhanced Threat Detection: By correlating data from different sources, SIEM systems can detect complex and multi-stage attacks that might be missed by individual security tools.

Improved Incident Response: SIEM provides security teams with the information needed to quickly identify and respond to security incidents.

Centralized Visibility: It offers a single pane of glass for monitoring and analysing security data across the entire network.

Regulatory Compliance: Helps organizations meet various regulatory requirements by maintaining and analysing logs and generating necessary reports.

Today we will build a SIEM in our own system

Prerequisites

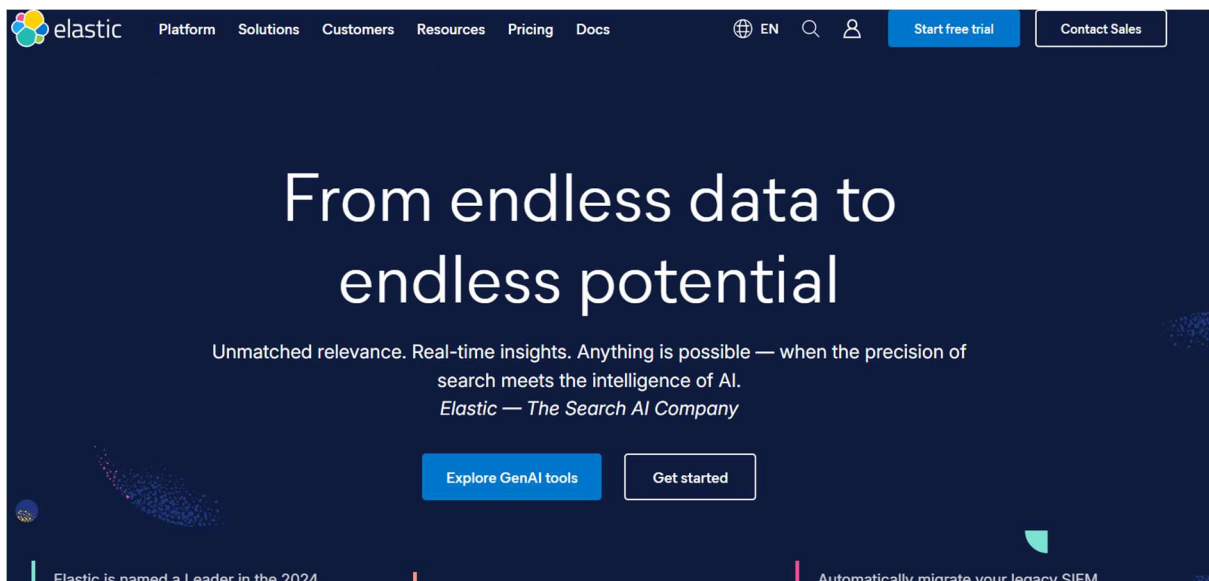
Before we get started, make sure you have the following:

1. VirtualBox or VMware
2. Basic knowledge of Linux and virtualization software.

Task 1: Set up an Elastic Account

Before we get started, we need to create a free account to set up a cloud Elastic instance that we can run the SIEM on. To do that, follow these steps:

1. Sign up for a free trial to use Elastic Cloud at <https://cloud.elastic.co/registration>
2. Once you have an Elastic account, log in to the Elastic Cloud console at <https://cloud.elastic.co>.
3. Click on “Start your free trial.”



What is Elastic.co used for?

Elastic.co is the official website for Elastic, the company behind Elasticsearch, Kibana, Beats, and Logstash, which are part of the Elastic Stack (formerly known as the ELK Stack). These tools are primarily used for searching, analysing, and visualizing data in real-time.

1. Elasticsearch: A highly scalable open-source search and analytics engine. It's used for full-text search, logging, monitoring, and more. Elasticsearch is the core of the Elastic Stack, allowing for fast search and data retrieval.

2. Kibana: A visualization tool that works with Elasticsearch. It allows users to create and share dynamic dashboards for visualizing data, making it easier to analyse trends and patterns.

3. Logstash: A data processing pipeline that ingests data from various sources, transforms it, and then sends it to a storage like Elasticsearch. It's often used for log and event data.

4. Beats: Lightweight data shippers that send data from various sources to Elasticsearch or Logstash. There are different types of Beats, such as File beat (for log files), Metric beat (for system metrics), and others tailored to specific data types.

Elastic.co provides these tools to help organizations make sense of large volumes of data, enabling powerful search capabilities, monitoring, and data analysis in various domains, including IT operations, security, and business analytics.

Task 2: Setting up the Linux VM

Next, we need to set up the Linux VM. You can use any Linux OS and virtualization software for this, but I'll be using Kali Linux and Oracle VirtualBox.

To set it up, follow these steps:

1. Download the Kali Linux VM from the official Kali website at <https://www.kali.org/get-kali/#kali-virtual-machines>.
2. Create a new VM with the Kali VM file in your preferred virtualization platform, such as VirtualBox or VMware.
3. Start the VM and follow the on-screen prompts to install Kali.
4. Once the installation is complete, log in to the Kali VM using the credentials "kali" for both the username and password.

The screenshot shows the Elastic UI interface. At the top, the Elastic logo is on the left, and a search bar with the text "Find apps, con" is on the right. Below the logo, a red box highlights the hamburger menu icon. To its right is a blue square icon with a white letter 'D'. Further right are two buttons: "Integrations" (highlighted with a red arrow) and "Browse integrations". Below these is a dark bar with a "Manage this deployment" button. The left sidebar contains a "Home" link and two main sections: "Analytics" and "Enterprise Search", each with a dropdown arrow. Under "Analytics" are links for Discover, Dashboard, Canvas, Maps, Machine Learning, Graph, and Visualize Library. Under "Enterprise Search" are links for Overview, Content, Behavioral Analytics, Elasticsearch, App Search, Workplace Search, and Search Experiences. At the bottom of the sidebar, the "Observability" section is highlighted with a red box, containing a blue "Add integrations" button. A red arrow points from the bottom right towards this button. The main content area is titled "Integrations" and has a close button (X). It features two tabs: "Integrations" and "Installed integrations". Below the tabs is a table with two columns. The first column lists integration names, and the second column shows counts in colored boxes. The counts are: 310 (pink), 2 (grey), 33 (grey), 23 (grey), 3 (grey), 5 (grey), 3 (grey), 15 (grey), 2 (grey), 27 (grey), 3 (grey), 1 (grey), and 35 (grey).

Integration	Count
	310
	2
	33
	23
Network	3
	5
	3
	15
agement	2
	27
	3
	1
	35

Task 3: Setting up the Agent to Collect Logs

An agent is a software program that is installed on a device, such as a server or endpoint, to collect and send data to a centralized system for analysis and monitoring. In the context of Elastic SIEM, an agent is used to collect and forward security-related events from your endpoints to your Elastic SIEM instance.

To set up the agent to collect logs from your Kali VM and forward them to your Elastic SIEM instance, follow these steps:

1. Log in to your Elastic SIEM instance and navigate to the Integrations page by: clicking on the Kibana main menu bar at the top left, then selecting “Integrations” at the bottom.
2. Search for “Elastic Defend” and click on it to open the integration page.

Integrations

Choose an integration to start collecting and analyzing your data.

[Browse integrations](#) **Installed integrations**

All categories

AuditD

AWS

Azure

Content Delivery Network

Cloud

310

2

33

23

3

5



Search for integrations



APM

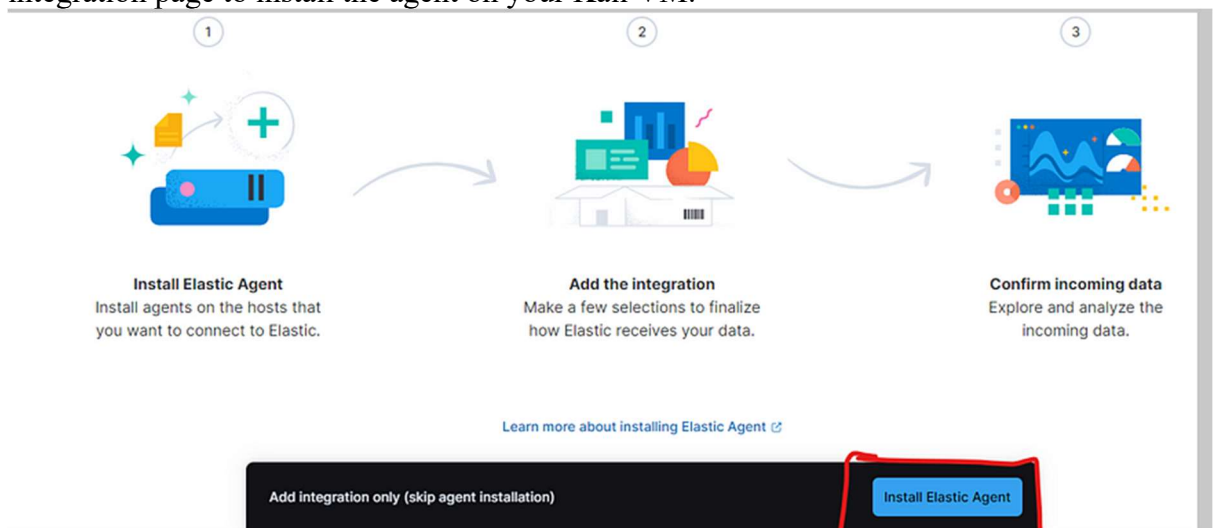
Collect performance metrics from your applications with Elastic APM.



Elastic Defend

Protect your hosts and cloud workloads with threat prevention, detection, and deep security data visibility.

Click on “Install Elastic Defend” and follow the instructions provided on the integration page to install the agent on your Kali VM.



These steps configure and enroll the Elastic Agent in Fleet to automatically deploy updates and centrally manage the agent. As an alternative to Fleet, advanced users can run agents in [standalone mode](#).

1 Install Elastic Agent on your host

Select the appropriate platform and run commands to install, enroll, and start Elastic Agent. Reuse commands to set up agents on more than one host. For aarch64, see our [downloads page](#). For additional guidance, see our [installation docs](#).

Linux Tar

Mac

Windows

RPM

DEB

Kubernetes

```
curl -L -O https://artifacts.elastic.co/downloads/beats/elastic-agent/elastic-agent-8.7.0-linux-x86_64.tar.gz
tar xzvf elastic-agent-8.7.0-linux-x86_64.tar.gz
cd elastic-agent-8.7.0-linux-x86_64
sudo ./elastic-agent install --url=https://687d5af1ce2b4a28a0304c6fbeb3c396.fleet.us-central1.gcp.cloud.es.io:443 --enrollment-token=eFBR5Xk0Y0JuQXg5M2YxVXc5VmM6czBqdEk3Y3VUX2VEaV90d0hmejNxQQ=
```

Copy to clipboard

Paste that command into the Kali terminal (command line).

```
(kali@kali)-[~/Desktop]
$ sudo curl -L -O https://artifacts.elastic.co/downloads/beats/elastic-agent/elastic-agent-8.7.0-linux-x86_64.tar.gz
tar xzvf elastic-agent-8.7.0-linux-x86_64.tar.gz
cd elastic-agent-8.7.0-linux-x86_64
sudo ./elastic-agent install --url=https://687d5af1ce2b4a28a0304c6fbeb3c396.fleet.us-central1.gcp.cloud.es.io:443 --enrollment-token=eFBR5Xk0Y0JuQXg5M2YxVXc5VmM6czBqdEk3Y3VUX2VEaV90d0hmejNxQQ=
[sudo] password for kali:
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
0         0     0    0     0      0     0      0  0:00:00  0:00:00  0:00:00  0
0         0     0    0     0      0     0      0  0:00:00  0:00:00  0:00:00  0
0  407M    0  346k    0     0   214k    0  0:02:20  0:00:01  0:02:19  214
0  407M    0 1695k    0     0   567k    0  0:01:14  0:00:02  0:01:12  567
```

Once the agent is installed, which can take a few minutes, you'll see a message that says "Elastic Agent has been successfully installed." It will automatically start collecting and forwarding logs to your Elastic SIEM instance, although it might take a few minutes for the logs to appear in the SIEM.

```
ent to URL: https://687d5af1ce2b4a28a0304c6fbeb3c396.fleet.us-central1.gcp.cloud.es.io:443/", "ecs.version": "1.6.0"}
{"log.level": "info", "@timestamp": "2023-04-29T14:42:49.650-0400", "log.origin": {"file.name": "cmd/enroll_cmd.go", "file.line": 273}, "message": "Successfully triggered restart on running Elastic Agent.", "ecs.version": "1.6.0"}
Successfully enrolled the Elastic Agent.
Elastic Agent has been successfully installed.
```

You can verify that the agent has been installed correctly by running this command: `sudo systemctl status elastic-agent.service`


```
└─$ sudo systemctl status elastic-agent.service
● elastic-agent.service - Elastic Agent is a unified agent to observe, monitor and pro>
   Loaded: loaded (/etc/systemd/system/elastic-agent.service; enabled; preset: disab>
   Active: active (running) since Sat 2023-04-29 14:42:46 EDT; 36min ago
     Main PID: 47316 (elastic-agent)
        Tasks: 41 (limit: 2271)
      Memory: 305.6M
         CPU: 10.835s
       CGroup: /system.slice/elastic-agent.service
              └─47316 elastic-agent
```

If you get an error installing the agent, make sure that your Kali is connected to the internet before proceeding by pinging google.com.

Task 4: Generating Security Events on the Kali VM

To verify that the agent is working correctly, you can generate some security-related events on your Kali VM. To do this, we can use a tool like Nmap. Nmap (Network Mapper) is a free and open-source utility used for network exploration, management, and security auditing. It is designed to discover hosts and services on a computer network, thus creating a “map” of the network. Nmap can be used to scan hosts for open ports, determine the operating system and software running on the target system, and gather other information about the network.

To run an Nmap scan, follow these steps:

1. Install Nmap on the Linux VM if you’re not using Kali, Nmap already comes preinstalled in Kali. Open a new Terminal and run this command to install it: `sudo apt-get install nmap`.
2. Run a scan on Kali machine by running the command: `sudo nmap <vm-ip>`. You can also run a scan of your host machine if you place your Kali VM on a “bridged” network.

```
(kali㉿kali)-[~]
└─$ sudo nmap 192.168.43.28
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-29 14:16 EDT
Nmap scan report for ANP (192.168.43.28)
Host is up (0.00034s latency).
All 1000 scanned ports on ANP (192.168.43.28) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: E4:B3:18:5B:D4:1B (Intel Corporate)
```

An Nmap scan of my host machine.

3. This scan generates several security events, such as the detection of open ports and the identification of services running on those ports. **Run a few more Nmap scans** (“`nmap -sS <ip address>`”, “`nmap -sT <ip address>`”, “`nmap -p- <ip address>`”etc..”

```

(kali㉿kali)-[~]
$ sudo nmap -A -p- 192.168.43.28
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-29 14:48 EDT
Nmap scan report for ANP (192.168.43.28)
Host is up (0.00036s latency).
Not shown: 65526 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
2070/tcp  closed ah-esp-encap
2072/tcp  closed msync
2074/tcp  closed vrtl-vmf-sa
3232/tcp  closed mdtp
3989/tcp  closed bv-queryengine
4267/tcp  closed vrml-multi-use
4269/tcp  closed vrml-multi-use
9831/tcp  closed unknown
11010/tcp closed unknown
MAC Address: E4:B3:18:5B:D4:1B (Intel Corporate)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   0.36 ms ANP (192.168.43.28)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 262.81 seconds

(kali㉿kali)-[~]
$ sudo nmap -sT 192.168.43.28
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-29 14:53 EDT
Nmap scan report for ANP (192.168.43.28)
Host is up (0.00020s latency).
All 1000 scanned ports on ANP (192.168.43.28) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: E4:B3:18:5B:D4:1B (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 23.49 seconds

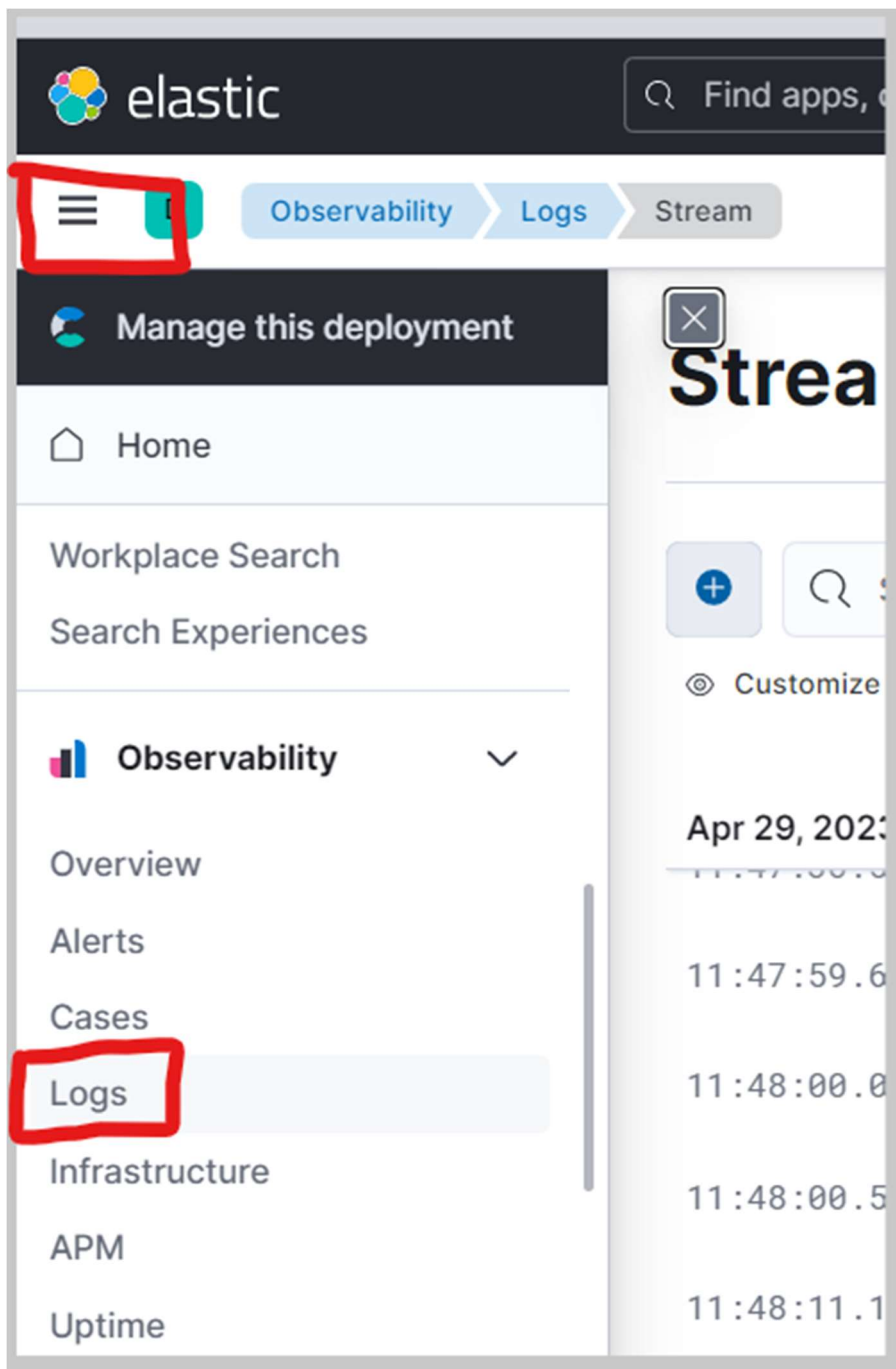
```

Task 5: Querying for Security Events in the Elastic SIEM

Now that we have forwarded data from the Kali VM to the SIEM, we can start querying and analyzing the logs in the SIEM.

To do this, follow these steps:

1. Inside your Elastic Deployment, click on the menu icon at the top-left with the three horizontal lines and then click on the “Logs” tab under “Observability” to view the logs from the Kali VM.



2. In the search bar, enter a search query to filter the logs. For example, to search for all logs related to Nmap scans, enter the query: `event.action: "nmap_scan"` or `process.args: "sudo"`.

3. Click on the "Search" button to execute the search query.

But please note that it can sometimes take a while for the events to populate and show up on the SIEM, so this query might not work right away.

4. The results of the search query will be displayed in the table below. You can click on the three dots next to each event to view more details.

Stream

process.args: "nmap"

CustomizeHighlights

Re

Apr 30, 2023	event.action	event.dataset	Message	
11:24:32.461	gid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:32.461	uid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:32.461	gid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:32.462	gid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:32.462	gid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:32.462	gid_change	endpoint.events.process	Endpoint process event	11:24 AM
11:24:34.288	uid_change	endpoint.events.process	Endpoint process event	11:24 PM

process.args

sudo, nmap, -p-, 10.0.2.15

process.args_count

4

process.command_line

sudo nmap -p- 10.0.2.15

process.command_line.caseless

sudo nmap -p- 10.0.2.15

process.command_line.text

sudo nmap -p- 10.0.2.15

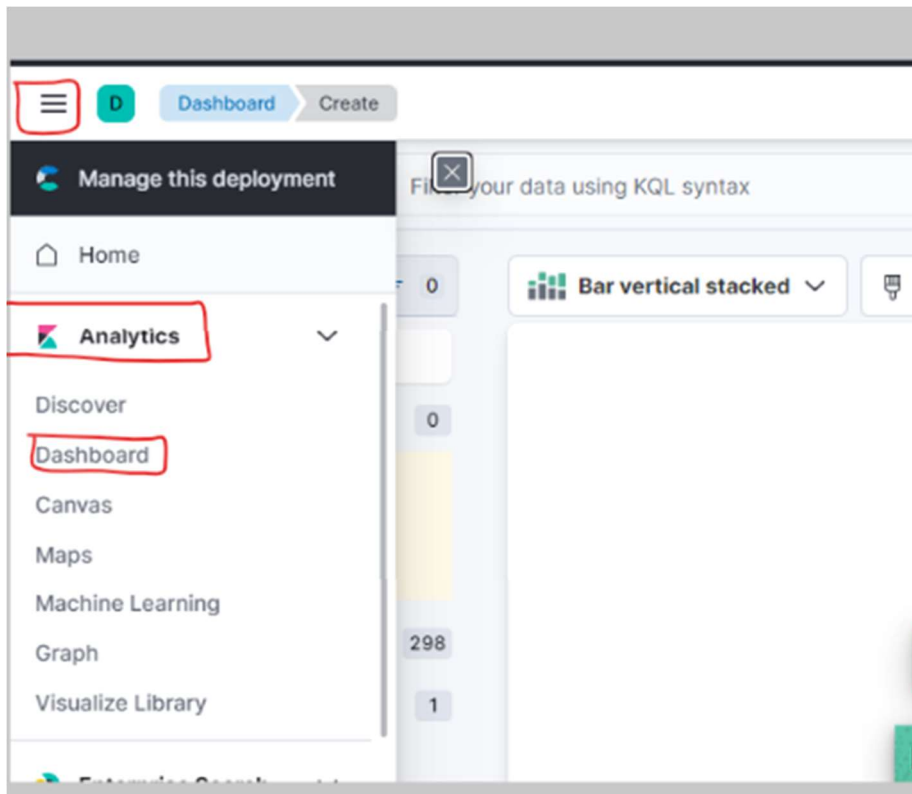
By generating and analyzing different types of security events in Elastic SIEM like the one above, or generating authentication failures by typing in the wrong password for a user or attempting SSH logins an incorrect password, you can gain a better understanding of how security incidents are detected, investigated, and responded to in real-world environments.

Task 6: Create a Dashboard to Visualize the Events

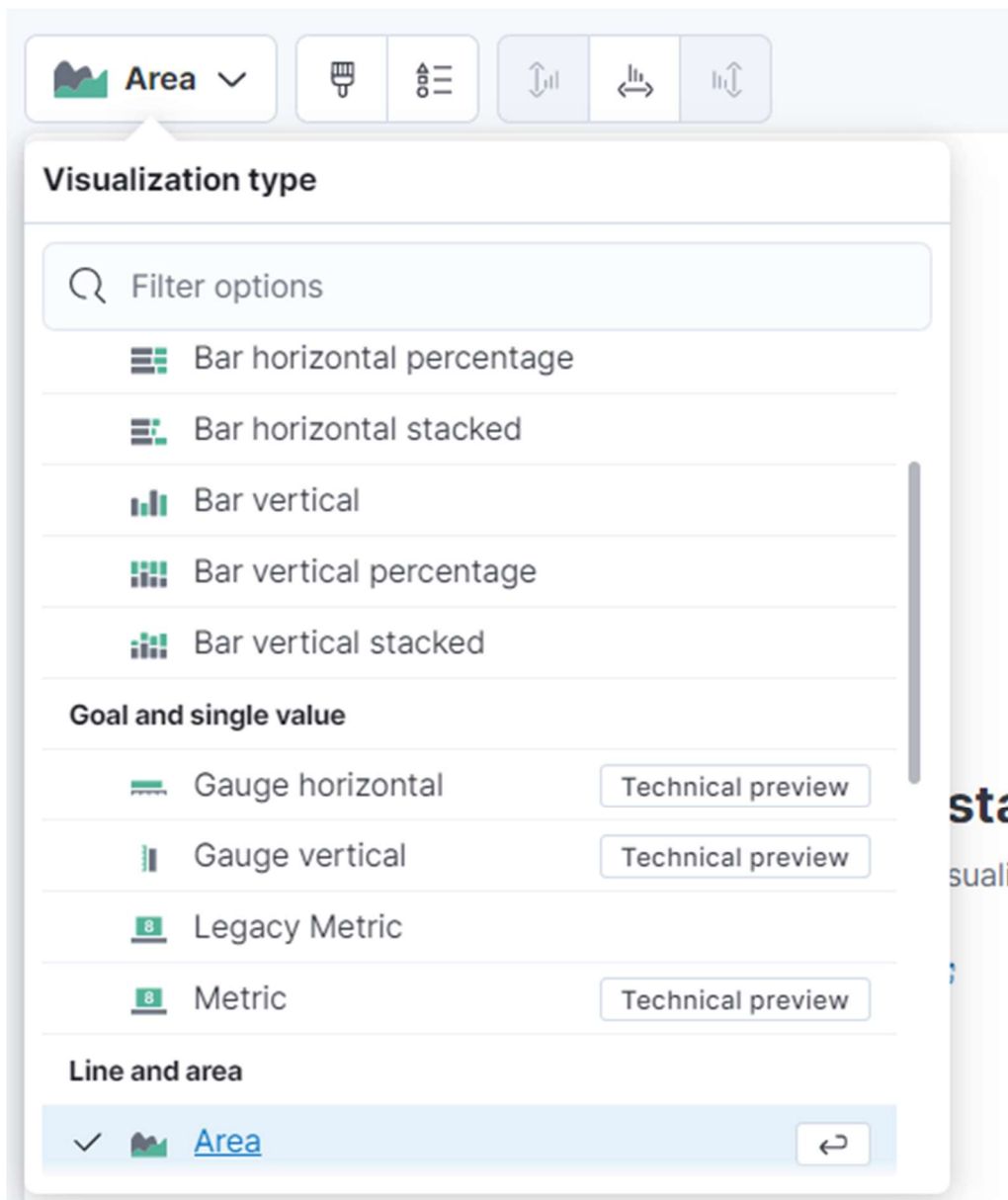
You can also use the visualizations and dashboards in the SIEM app to analyze the logs and identify patterns or anomalies in the data. For example, you can create a simple dashboard that shows a count of security events over time.

Here's how you can do that:

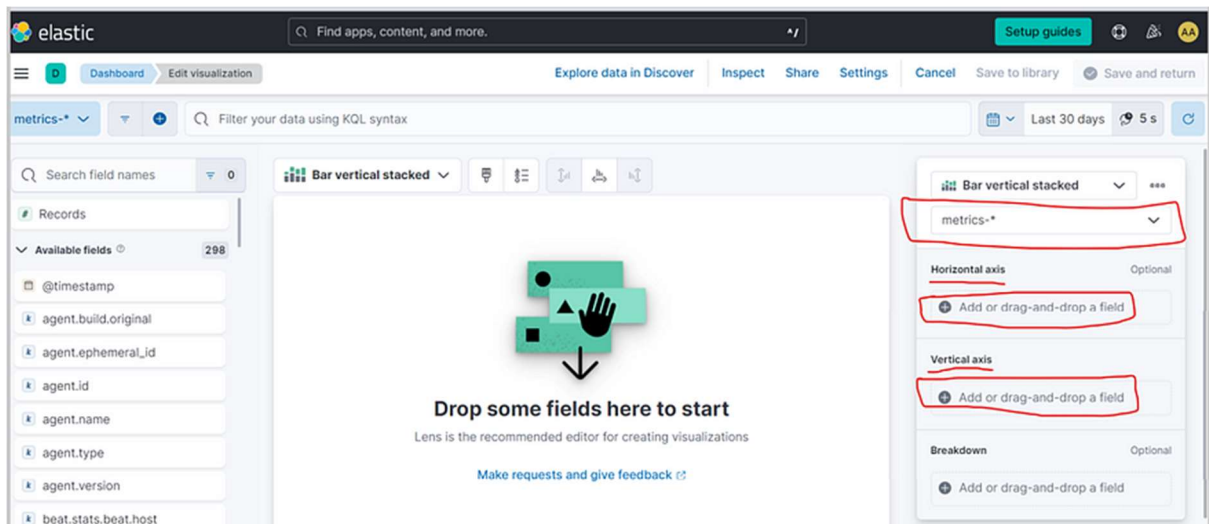
1. Navigate to the Elastic web portal at <https://cloud.elastic.co/>.
2. Click on the menu icon on the top-left, then under "Analytics," click on "Dashboards."



3. Click on the “Create dashboard” button on the top right to create a new dashboard.
4. Click on the “Create Visualization” button to add a new visualization to the dashboard.
4. Select “Area” or “Line” as the visualization type, depending on your preference. This will create a chart that shows the count of events over time.



6. In the “Metrics” section of the visualization editor on the right, select “Count” as the vertical field type and “Timestamp” for the horizontal field. This will show the count of events over time.



Horizontal axis

Data

Functions 

Date histogr...

Filters

Intervals •

Top values •

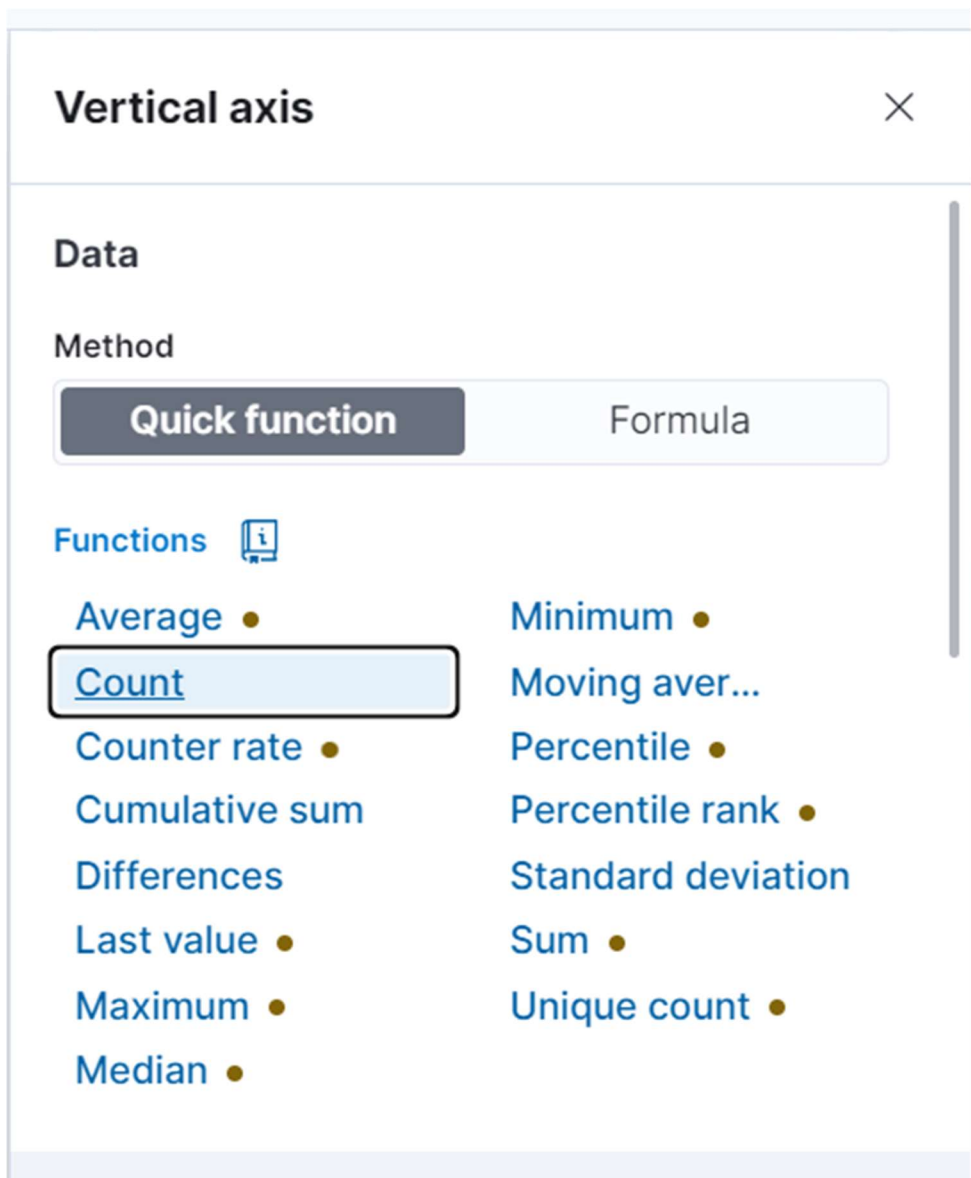
Field

@timestamp

☒ Include empty rows

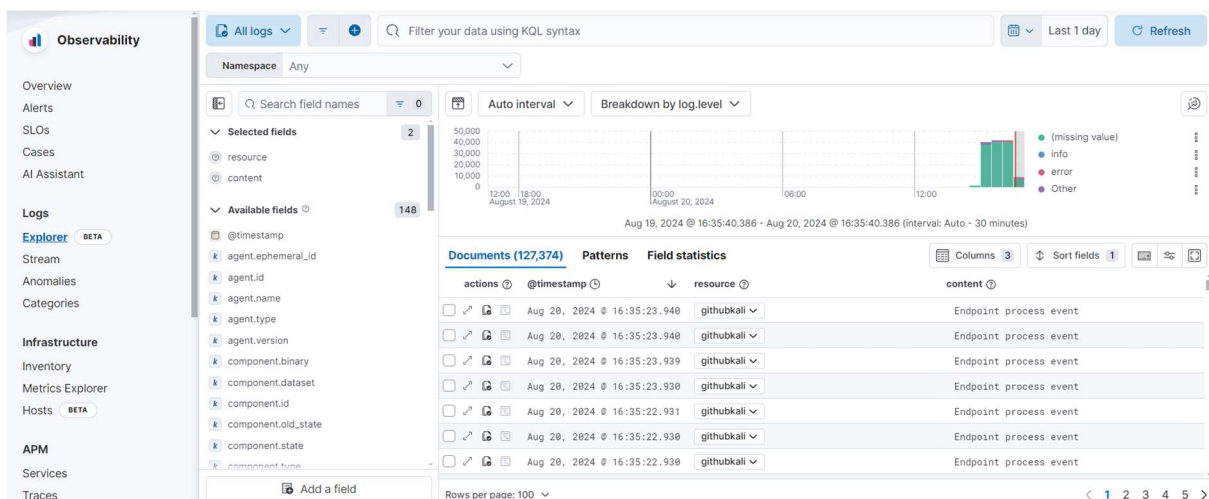
Minimum interval

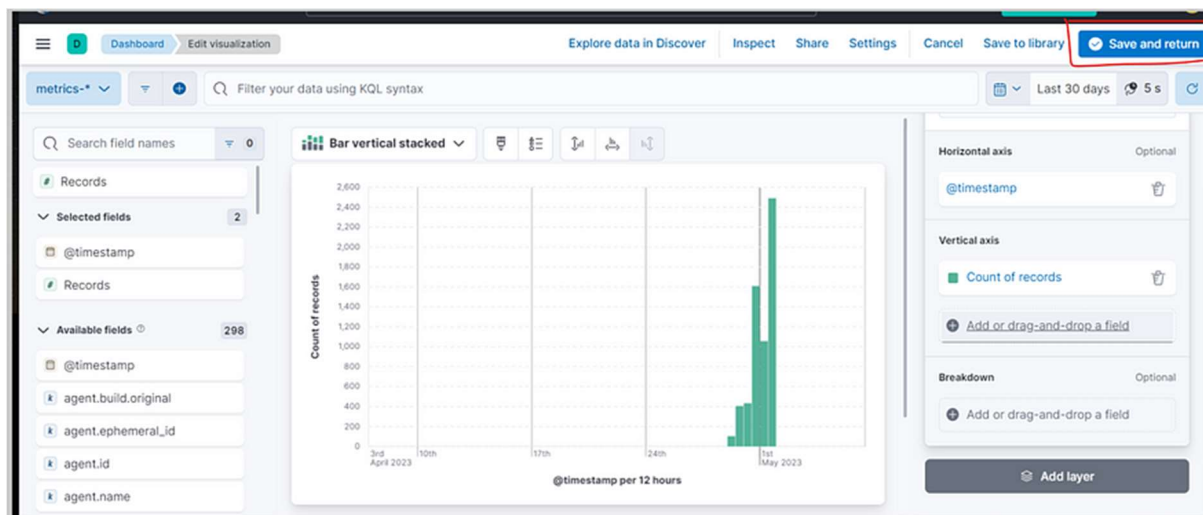
Click “close” once you’re done.



Click “close” once you’re done.

7. Click on the “Save” button to save the visualization and then complete the rest of the settings.





Conclusion

In this guide, we have set up a home lab using Elastic SIEM and a Kali VM. We forwarded data from the Kali VM to the SIEM using the Elastic Beats agent, generated security events on the Kali VM using Nmap, and queried and analysed the logs in the SIEM using the Elastic web interface. We also created a dashboard to visualize security events and then created an alert to detect security events.

This home lab provides a valuable environment for learning and practicing the skills necessary for effective security monitoring and incident response using Elastic SIEM. By following these steps, you can gain hands-on experience with using a SIEM and improve your security monitoring skills to help you become a successful security analyst or engineer.